

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

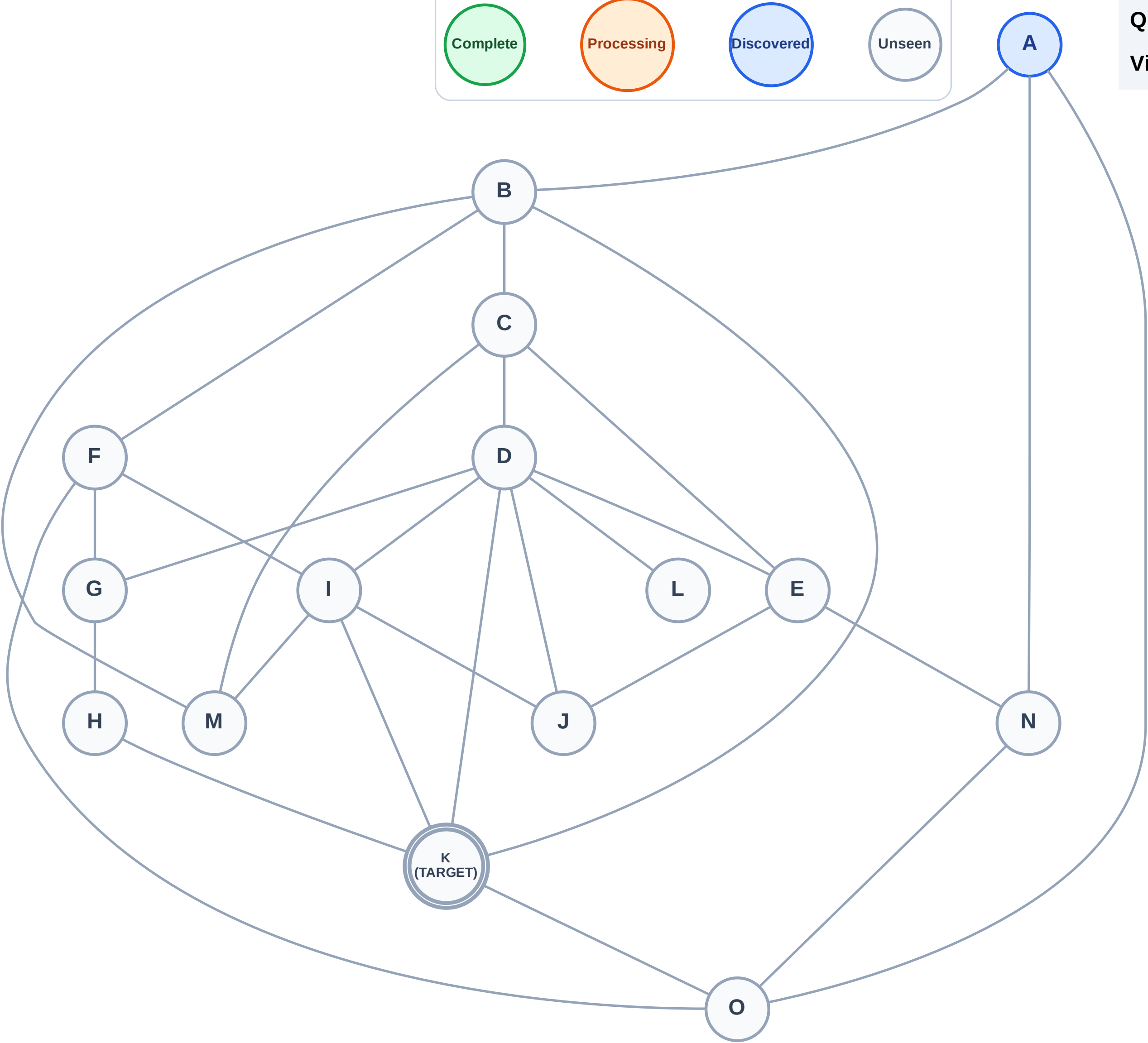
Complete

Processing

Discovered

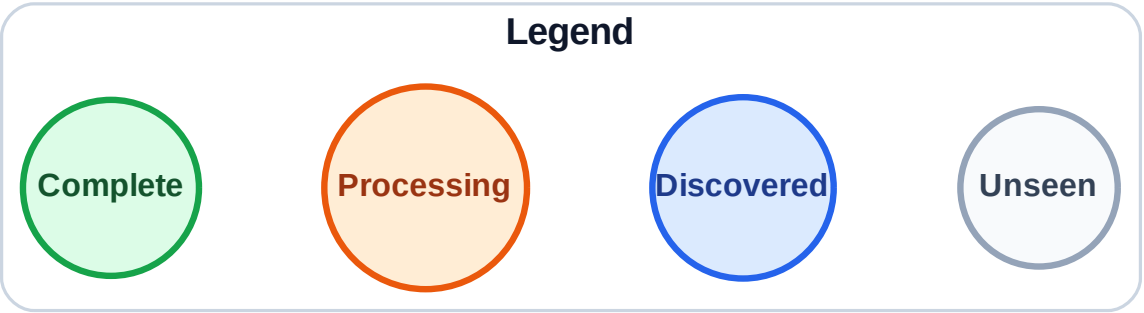
Unseen

Queue: A
Visited: (none)

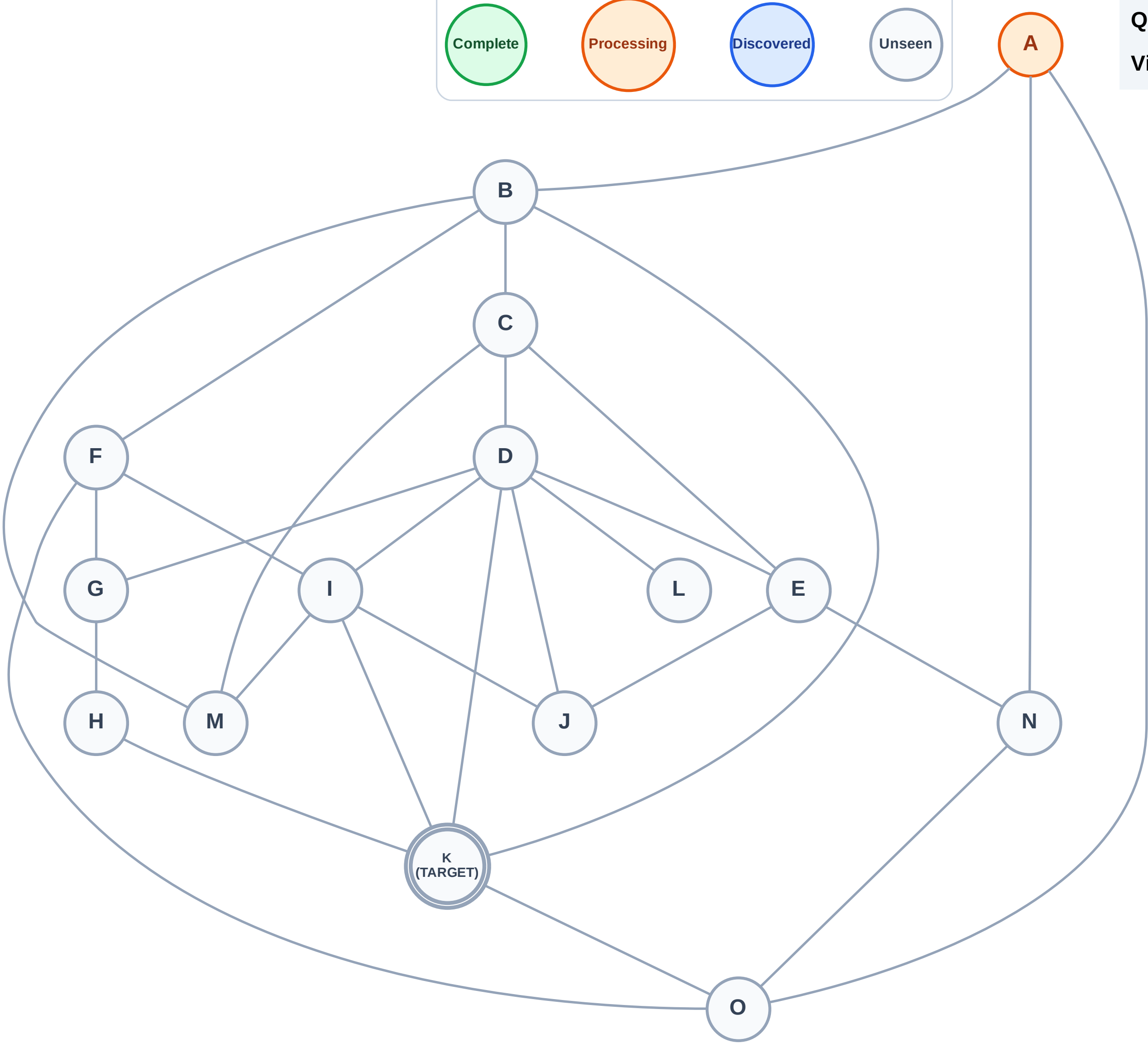


BFS Traversal

Step 2: Process node A



Queue: (empty)
Visited: (none)



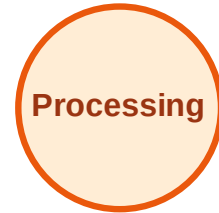
BFS Traversal

Step 3: Discover B, N, O from A

Legend



Complete



Processing

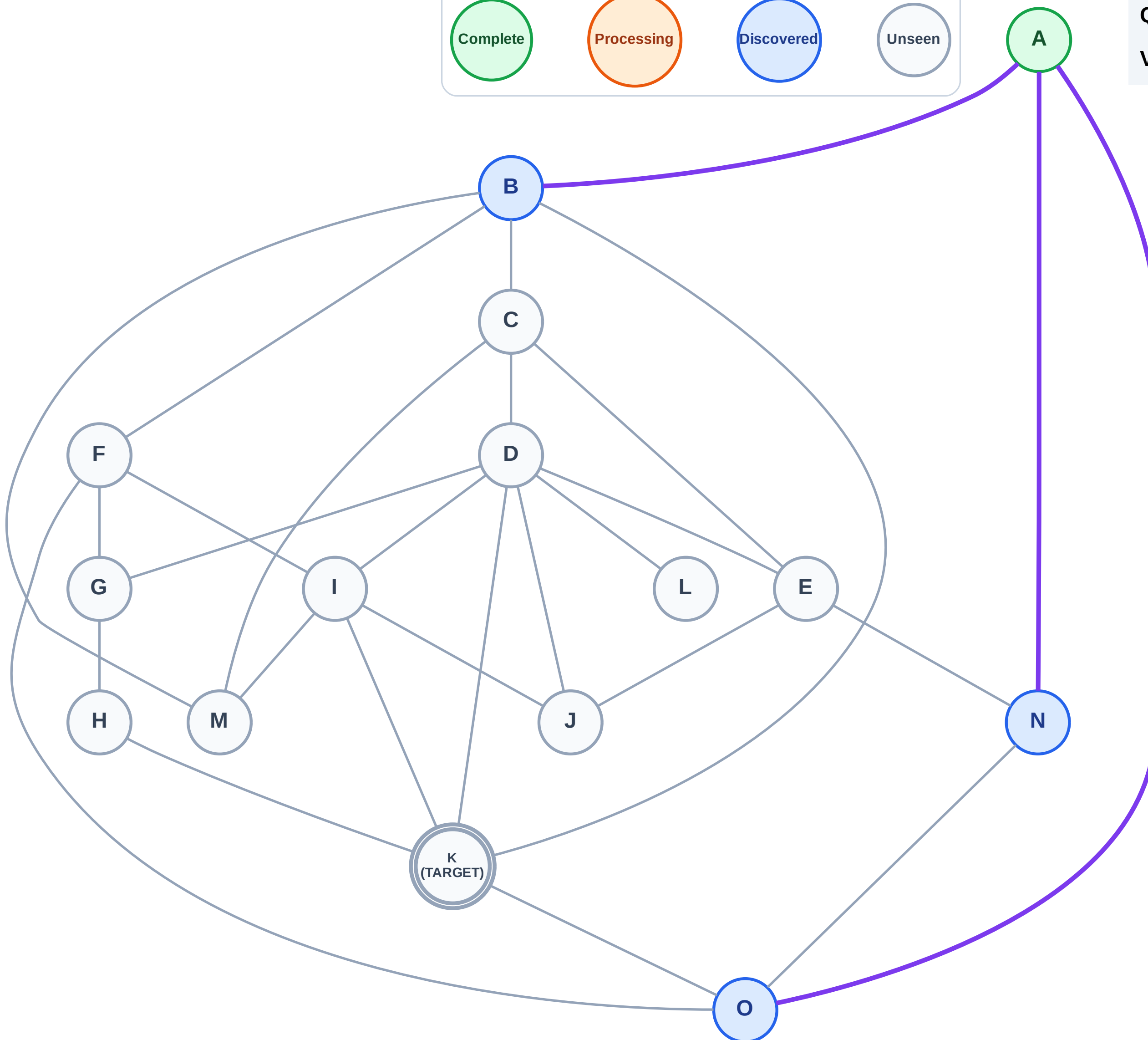


Discovered



Unseen

Queue: B → N → O
Visited: A



BFS Traversal

Step 4: Process node B

Legend

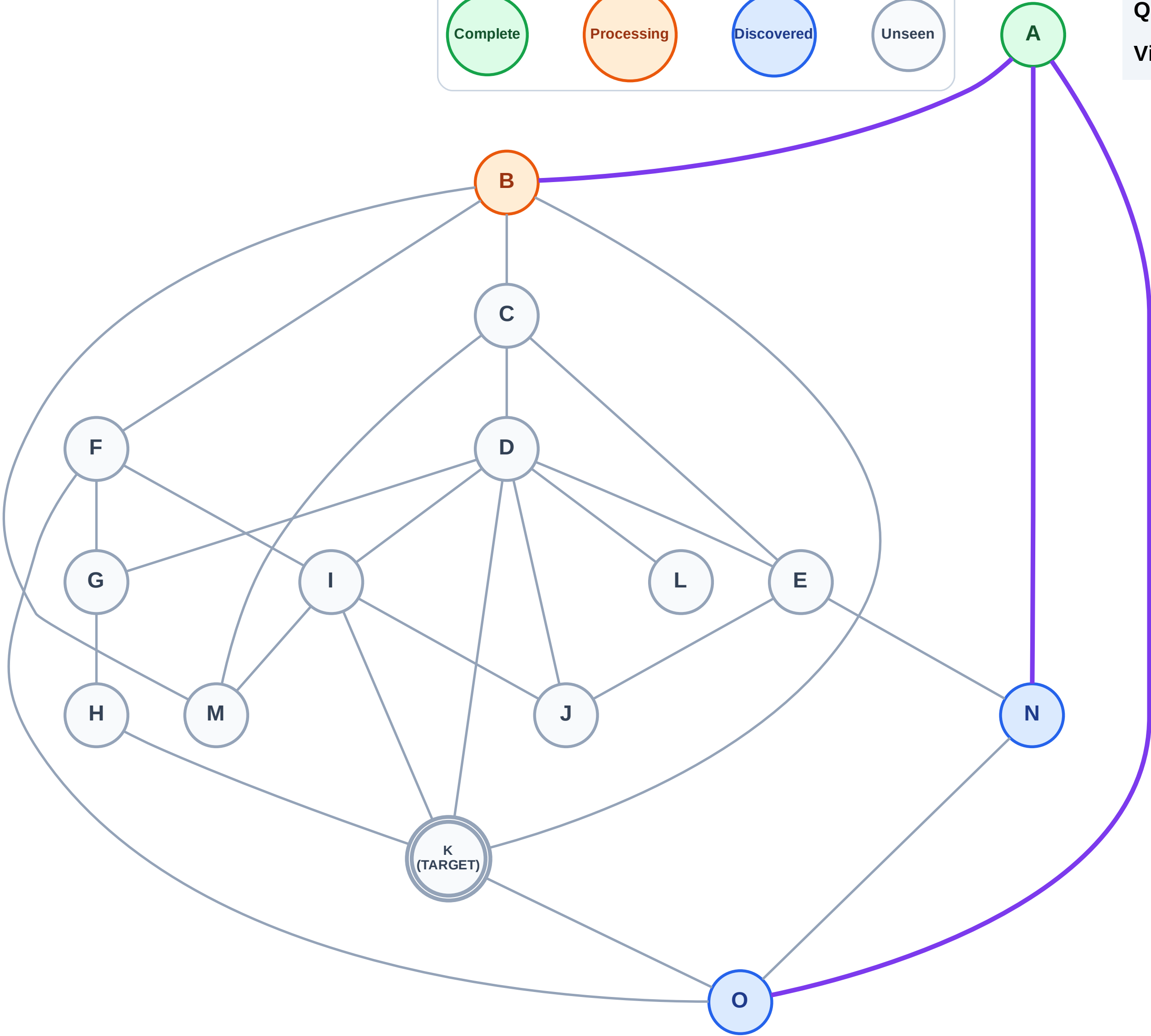
Complete

Processing

Discovered

Unseen

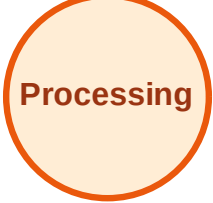
Queue: N → O
Visited: A



BFS Traversal

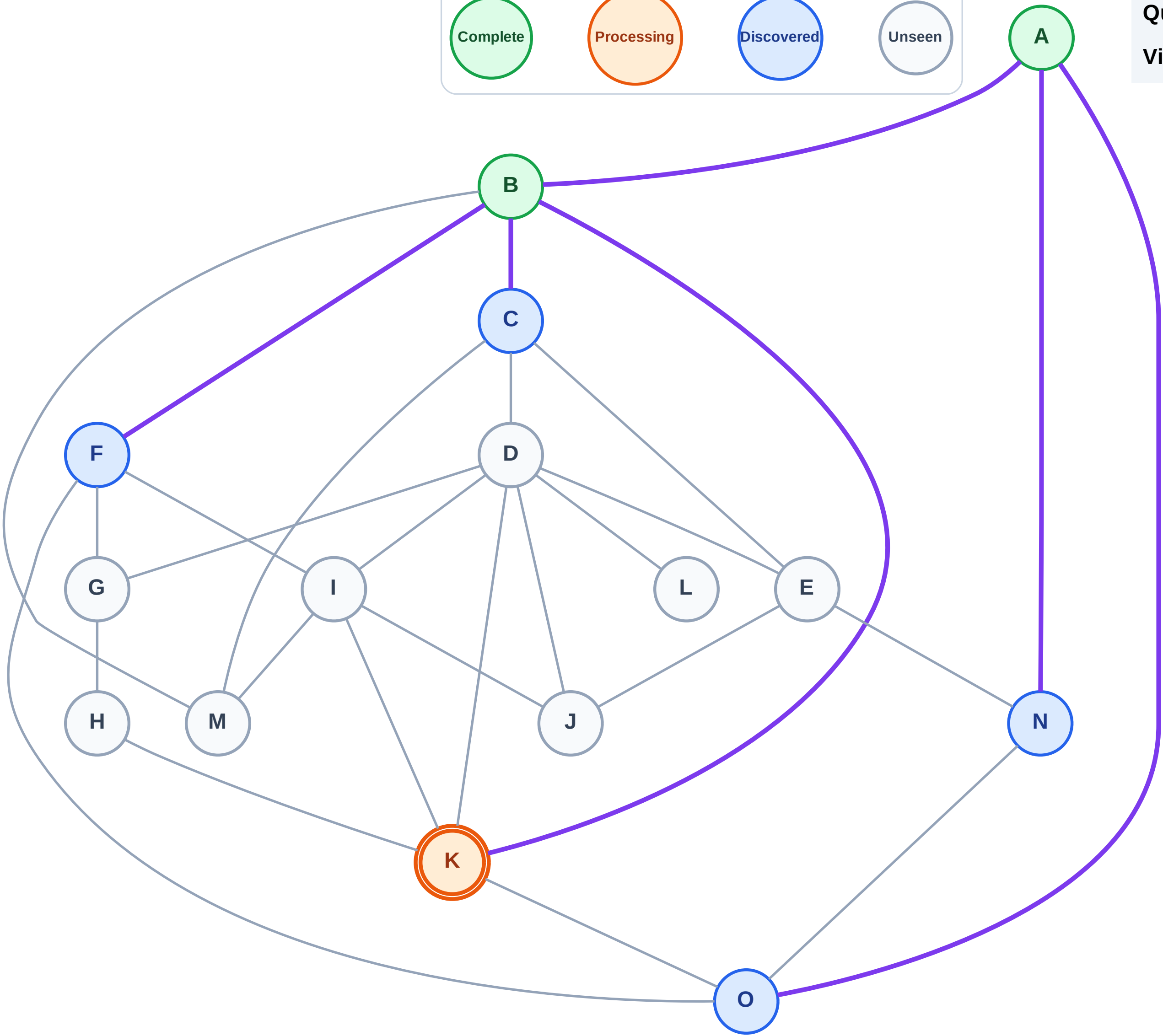
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: N → O → C → F

Visited: A, B, K



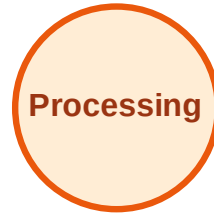
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

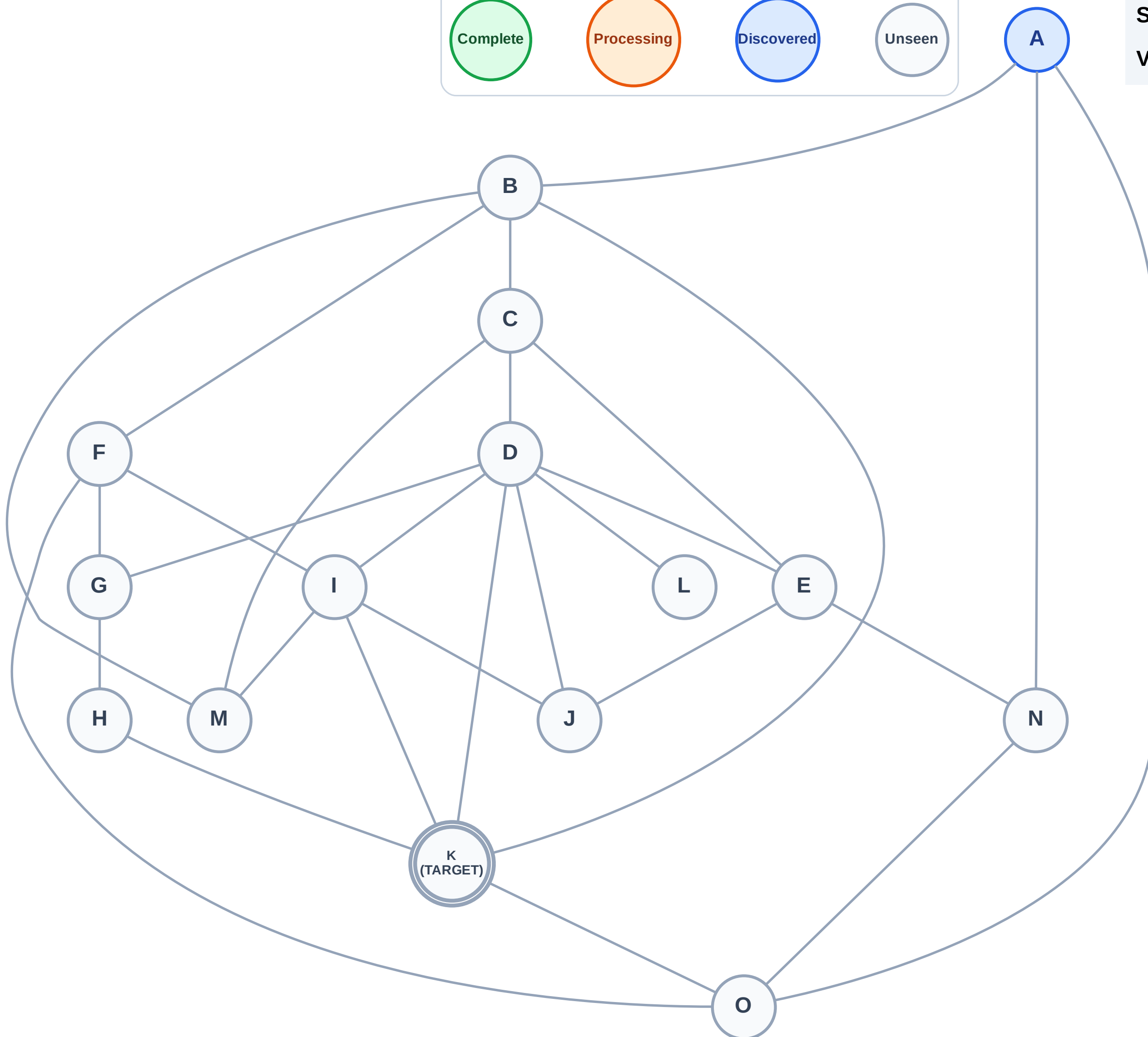


Discovered



Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

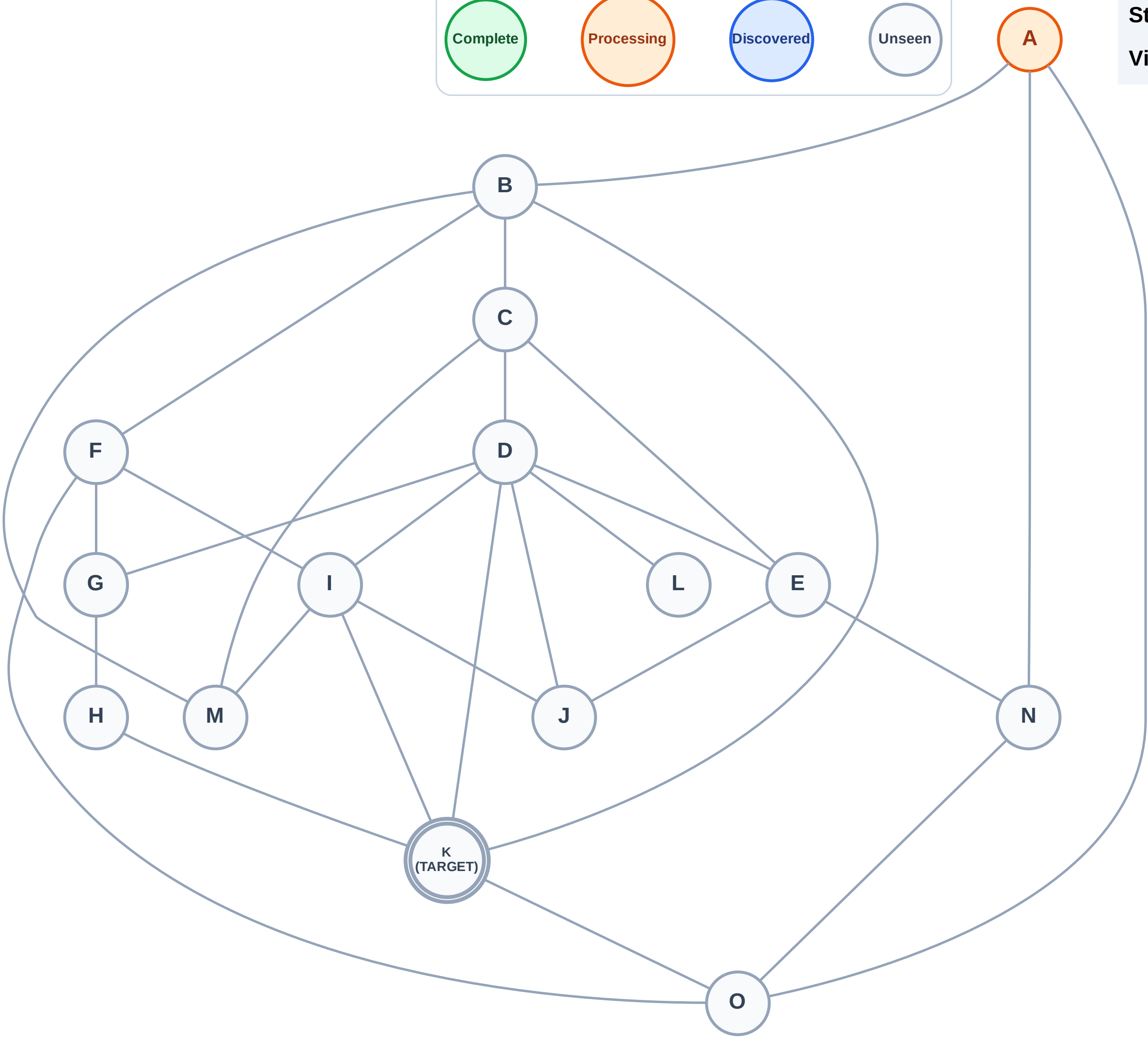
Step 2: Process node A

Legend



Stack (top at right): (empty)

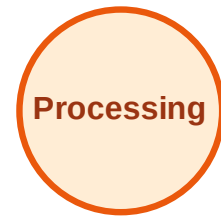
Visited: (none)



DFS Traversal

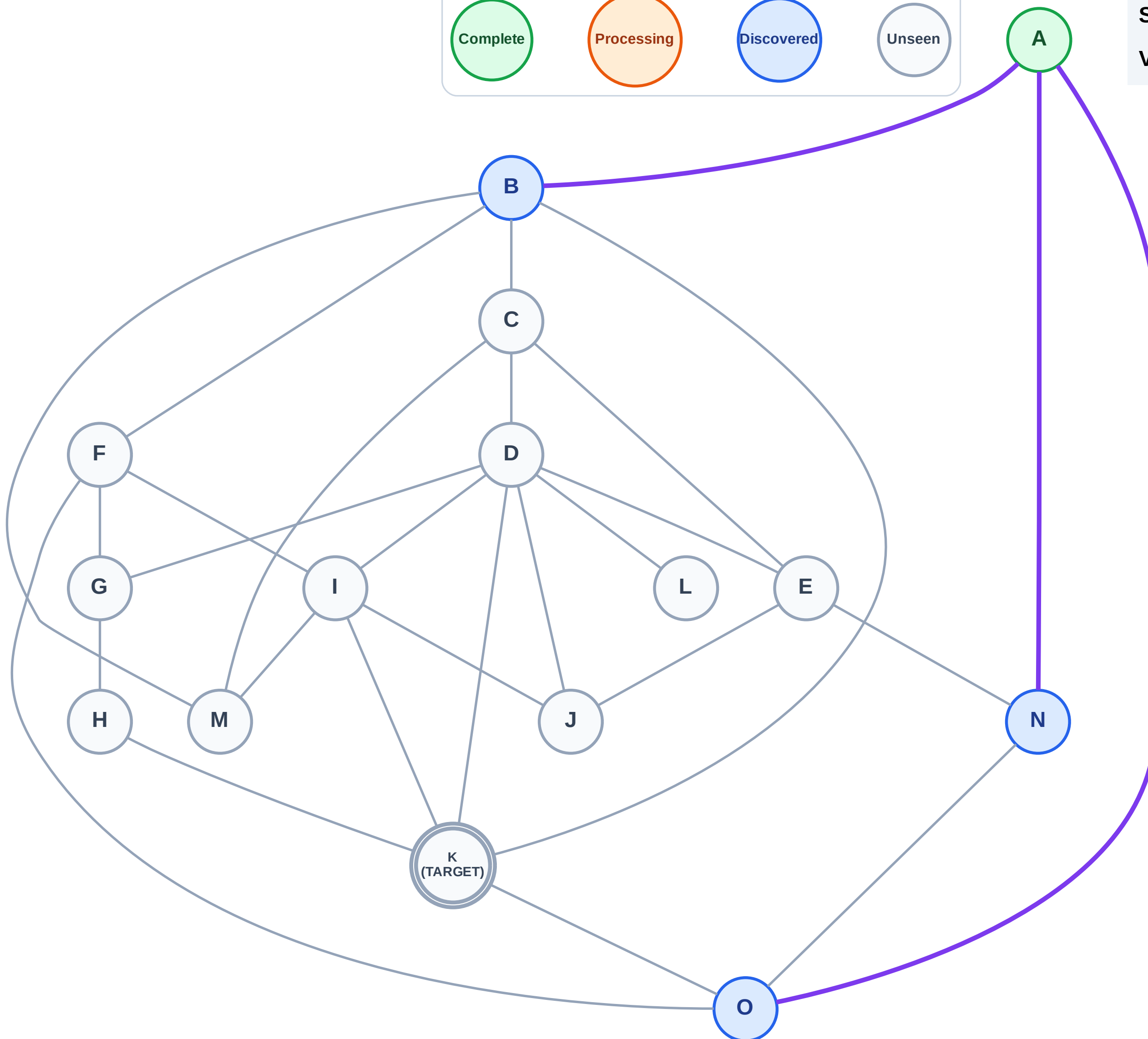
Step 3: Discover O, N, B from A

Legend



Stack (top at right): O | N | B

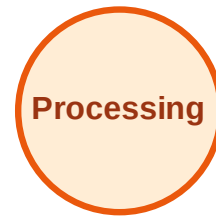
Visited: A



DFS Traversal

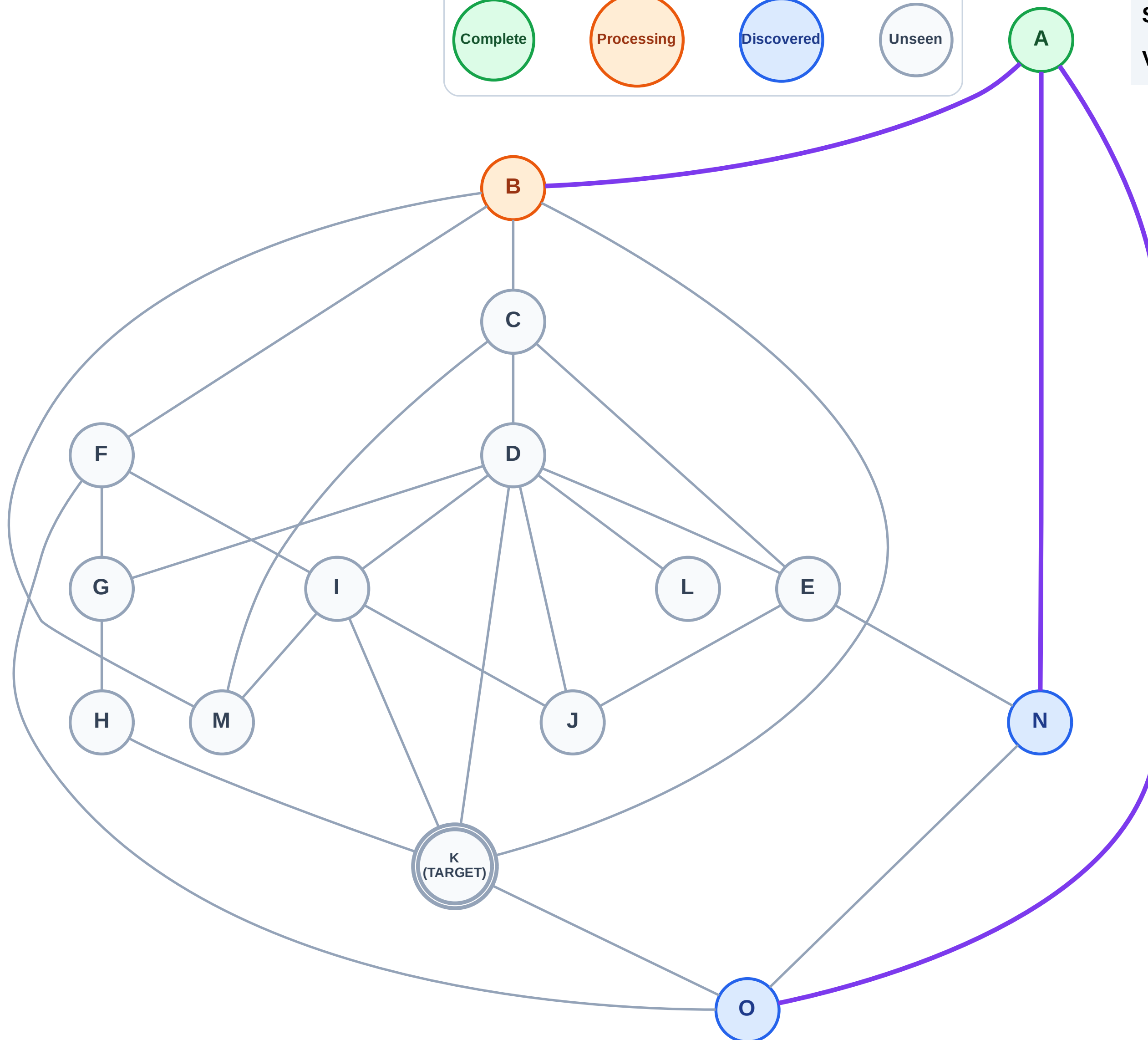
Step 4: Process node B

Legend



Stack (top at right): O | N

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M

Visited: A, B, K

